

WASP-12b

R-2184

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_150114 · created 2026-08-13T23:07:32+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457035.61456 +/- 0.00012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.045
Transit depth (Rp/Rs)^2	2.36 +/- 1.37 [%]
Orbital Inclination (inc)	86.73 +/- 2.9
Airmass coefficient 1 (a1)	0.64 +/- 0.3
Airmass coefficient 2 (a2)	0.29 +/- 0.25
Scatter in the residuals of the lightcurve fit is	43.21 %
Best Comparison Star	#3 - [165, 152]
Optimal Aperture	0.87
Optimal Annulus	3.5
Transit Duration (day)	0.0888 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.045 vs 0.1178 (deviation 0.64σ)
Duration fit vs published	0.0888 d vs 0.125 d (deviation 9.02σ)
Mid-time O–C	-1571.6 min
Depth SNR	2.7
Residual scatter	43.21 %

Light curve

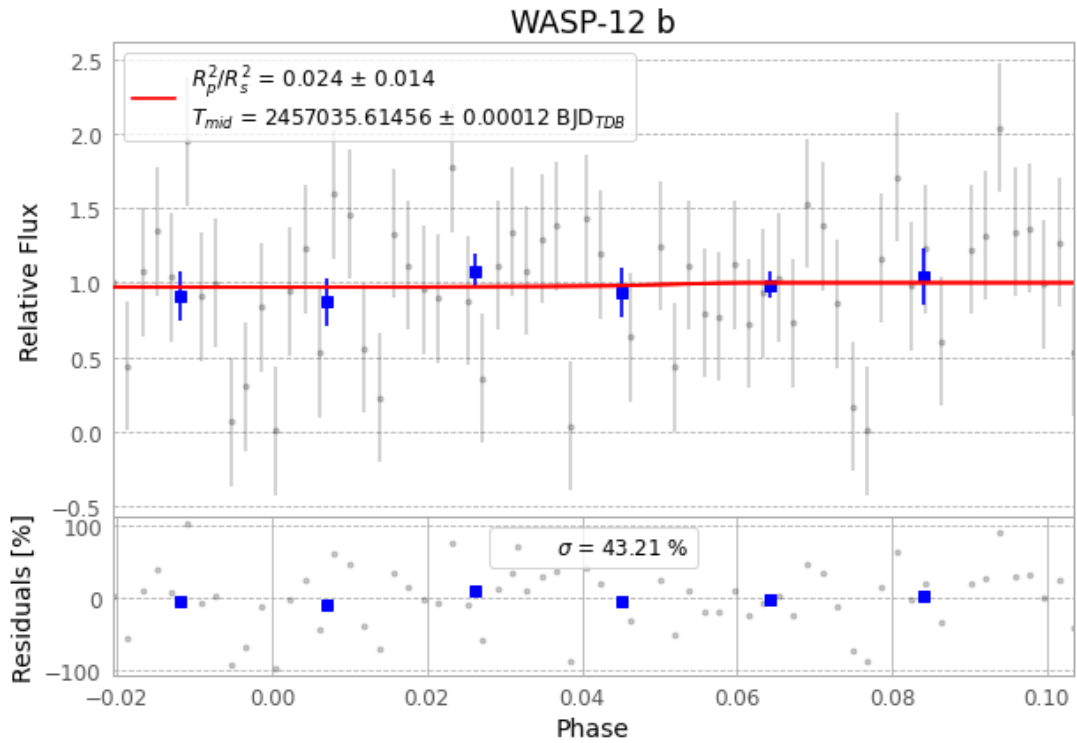


Figure 1 — FinalLightCurve_WASP-12 b_2015-01-12.png (SHA-256 886533a6bc1b0308...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [269, 237], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 fd2ed8f6fe258171...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

70 FITS frames (46 MB), WASP-12150113020312.FITS → WASP-12150113053012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-150114.log (31aa4b14c1c8...), FinalLightCurve_WASP-12 b_2015-01-12.png (886533a6bc1b...), FinalLightCurve_WASP-12 b_2015-01-12.csv (0cd3ebac8b83...)

Compute resources

Wall time 102.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 23:07Z.